

Project Briefs

A few project ideas have been given below. You may select any of these projects or similar ones. Your skillsets must match the project requirements, with an expectation that you should be able to execute the project independently. Our expectation of an 800-level project is that you will work independently on the project and apply professional judgement to come to a proper solution. Your overall level of achievement will be determined by the marks awarded (within specified percentage ranges), based on which a grade will be calculated.

Non-industry projects:

A. Develop a report that compares different websites from a user experience point of view, looking at the features and functionality offered, the information that can be shared and the design interface, and identify features that work well or don't work well. The websites should be in English language and could relate to any one of the following or similar others:

1. Real-estate company websites evaluation (Barfoot & Thompson, Harcourts, Ray White)
2. Hospitality sector websites evaluation (Copthorne Hotel, Sofitel Hotel, InterContinental Hotel)
3. Health sector websites evaluation (Auckland City Hospital, North Shore Hospital, Waitakere Hospital).
4. Travel sector websites evaluation (House of Travel, Flight Centre, Eco Travels)

B. Investigate the benefits and challenges in the use of ChatGPT based on functionality, technology, and the usability for different type of users (academia, industry practitioners, professionals). This can also be extended to comparison with DeepSeek.

C. Evaluate Sustainable Development Goals (SDGs) dataset by United Nations (<https://dashboards.sdgindex.org/downloads>) – Identify correlations between some of the existing 17 SDGs and their dimensions based on economic well-being, social justice and environmental sustainability ranked for 193 countries. The dataset shows the individual and cumulative score of different dimensions, overall trends since 2006, regional grouping, etc. The SDGs are complementary and, in some way, impact each other. Ask questions from the dataset, conduct relevant data analyses, and provide statistical evidence for your findings, backed by literature (e.g., which OECD countries show a downward trend in identified SDGs or contribute to low targets? This can be extended to other groupings (Oceania, Latin America, etc.)). Likewise, you can ask more questions and conduct new investigations.

D. Explore the intersection of geospatial intelligence, natural language processing (NLP), and user-centred web design through three applied projects aimed at enhancing how people interact with urban data.

1. Explainable Place Sentiment Map from User Reviews focuses on extracting and visualising emotional patterns tied to specific locations using sentiment and aspect-based opinion mining, paired with explainable AI techniques to increase trust and transparency.
2. Geospatial Question Answering Interface for Urban Data enables users to pose natural language queries about city issues, such as noise complaints or service distribution, and receive intuitive, map-based answers, making urban data more accessible and interactive.
3. Geo-located NLP Feedback System for Public Transport processes unstructured commuter feedback to classify concerns, detect spatial trends, and deliver insights through a real-time visual dashboard for transport planners. Together, these projects demonstrate how spatially aware NLP systems can improve civic engagement, decision-making, and transparency in smart cities.

Industry projects:

- 1- EquityAI: **Infracast** is a 24/7 geospatial analytics tool developed by EquityAI to support decision-making across transportation, environmental impact, housing economics, and infrastructure investment. It processes parsed, anonymised telecom data to derive real-time and forecasted insights on population movement (Infracast Move), emissions (Infracast Enviro), land and housing values (Infracast Econ), and asset valuations (Infracast Invest). The system integrates spatial data with demographic and temporal variables to enable high-resolution analysis down to individual roads or properties. A web-based user interface, built using JavaScript, Node.js, PostgreSQL, and Mapbox, displays origin-destination data, trip metrics, and financial outcomes across spatial zones.

The current phase of development extends the existing Mapbox-based UI by introducing new features: congestion detection and prediction using traffic elasticity models; multizone selection to analyse complex travel patterns; and route-based views tied to specific origin/destination pairs. These enhancements allow users to simulate traffic volume changes, visualise real-time congestion effects, and analyse network capacity at granular levels. The project uses dynamic floating panels, sliders, and charts to adjust and reflect user-defined scenarios, all powered by data stored in PostgreSQL and processed via EquityAI's backend infrastructure.

2- MasseyGowns: We use our website for graduates and staff (if required) to create an order for the academic regalia they will wear at their graduation ceremony. It involves the identification of their height, head size and the qualification they are graduating with. They include an amount of identification information. They are then taken to the Verifone (EFTPOS NZ) payment options for Credit/Debit card, Purchase Order Number or an on-line payment option. We do about 10,000 hires per year and about 50% are through the website. It also has several information pages about us, and the Graduate Women Manawatu Charitable Trust that owns us – Academic Dress Hire.

The website also has a regalia Buy option for individual items, or for sets. There is a Casual Hire for Photos option for those eligible to wear the regalia but are not attending the graduation ceremonies.

We do a data extract on a regular basis to upload the order information into a MS Access database (through a conversion spreadsheet for some changes, i.e. upper case surname). This database produces documents for the order preparation processes and the cost information is used for audit requirements. The extract is also uploaded through a conversion spreadsheet onto the Xero invoice template to create individual invoices in our accounting software - Xero. The database is also used to do a manual bulk upload of an organisation's orders, e.g. Massey Speaker orders, school orders (for prizegiving), smaller institutions like Pacific International Hotel Management School (PIHMS) in Taranaki.

Current Issues

1. We can edit informational aspects, change prices and add new regalia items (with difficulty) but the underlying software incurs a development cost from Zeald if we wish to make changes.
2. When we create (activate) the dormant page for a graduation, the dates etc are changed manually in multiple places and at times can miss some. This is currently for 8x graduations and could grow with another 4x graduation.
3. The website also has every combination of gown and trencher by size and style, plus the trencher/bonnet options, and then the numerous qualifications represented by a hood or stole. A one place data entry for changes is not available.
4. The data extract requires a conversion to gain some extra utility with the data in our database and accounting software.

Value Add Opportunity

There are several opportunities that can add value to this system:

- Producing the documentation that we use for preparing orders – order Master List by customer/graduation showing name, gown/hood/hat, Gown/Hood/Hat numbers by size required for an event, despatch documents, customer courier address label, download of information for interrogation/reporting.
- Allow us or customer organisations to upload bulk orders from a spreadsheet
- Producing a data extract in a format that can be directly copied into the XERO invoice template
- Produced a data extract that can be uploaded into an RFID tracking software solution

Website: www.masseygowns.org.nz